

The Problem

Company

Logistics Supply Chain Management

Context

A fashion and beauty startup company is experiencing an increase in major defects across product lines that exceed Acceptance Quality Limit (AQL) thresholds of 2.0%. Leadership wants to understand the operational impact of these defects.

Problem Statement

Review the inventory and shipping records for all product SKUs to determine the following:

- 1) What is the operational impact of quality control failures?
- 2) Which product lines carry the highest defect liability?
- 3) Which suppliers present the greatest fulfillment bottleneck?

The Process

Step 1

Create a project design document

- Identify what data sources are available to answer the business questions
- Review the integrity, quality, and limitations of the data

Tools used: Word, Excel

Step 2

Clean and query the raw data set

- Clean and query inventory/shipping records for all SKUs
- Perform complex queries to calculate relevant metrics

Tools used: SQL

Step 3

Create visualizations for insights

- Create visualizations that answer the business questions
- Use visualizations to construct a dashboard

Tools used: Tableau Public

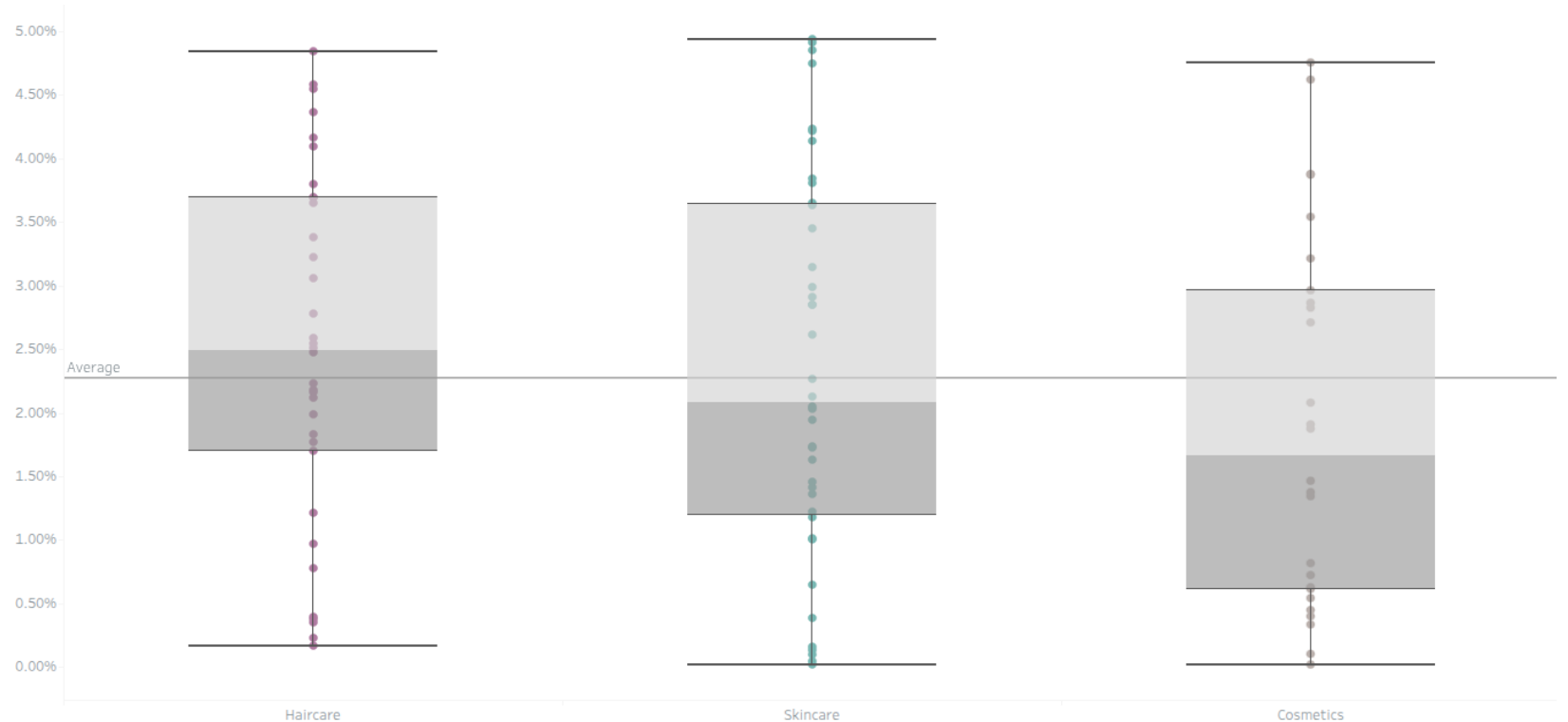
The Results

Average defect rate is 2.28% across all product lines

Insight:

The average defect rate is 2.28% across all product lines.

With 36% of batches failing inspection and 41% pending review, only 23% of products have passed inspection.



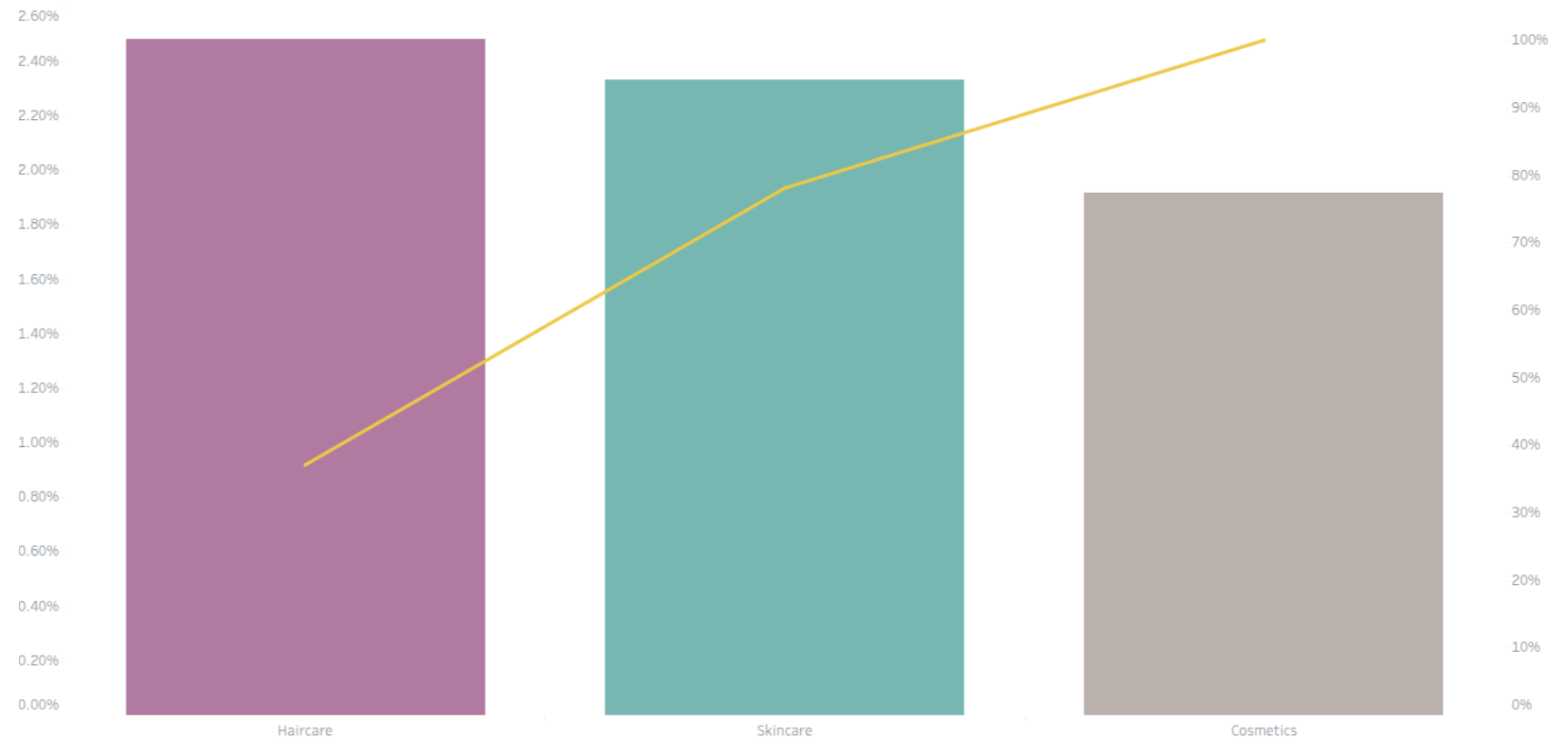
The Results

Haircare and Skincare lines require stricter quality control standards

Insight:

Haircare (2.48% average defect rate) and Skincare (2.33% average defect rate) exceed the 2.0% AQL threshold and require stricter quality control standards relative to Cosmetics (1.92% average defect rate).

Focusing on Haircare and Skincare will address 78.09% of the total defects.

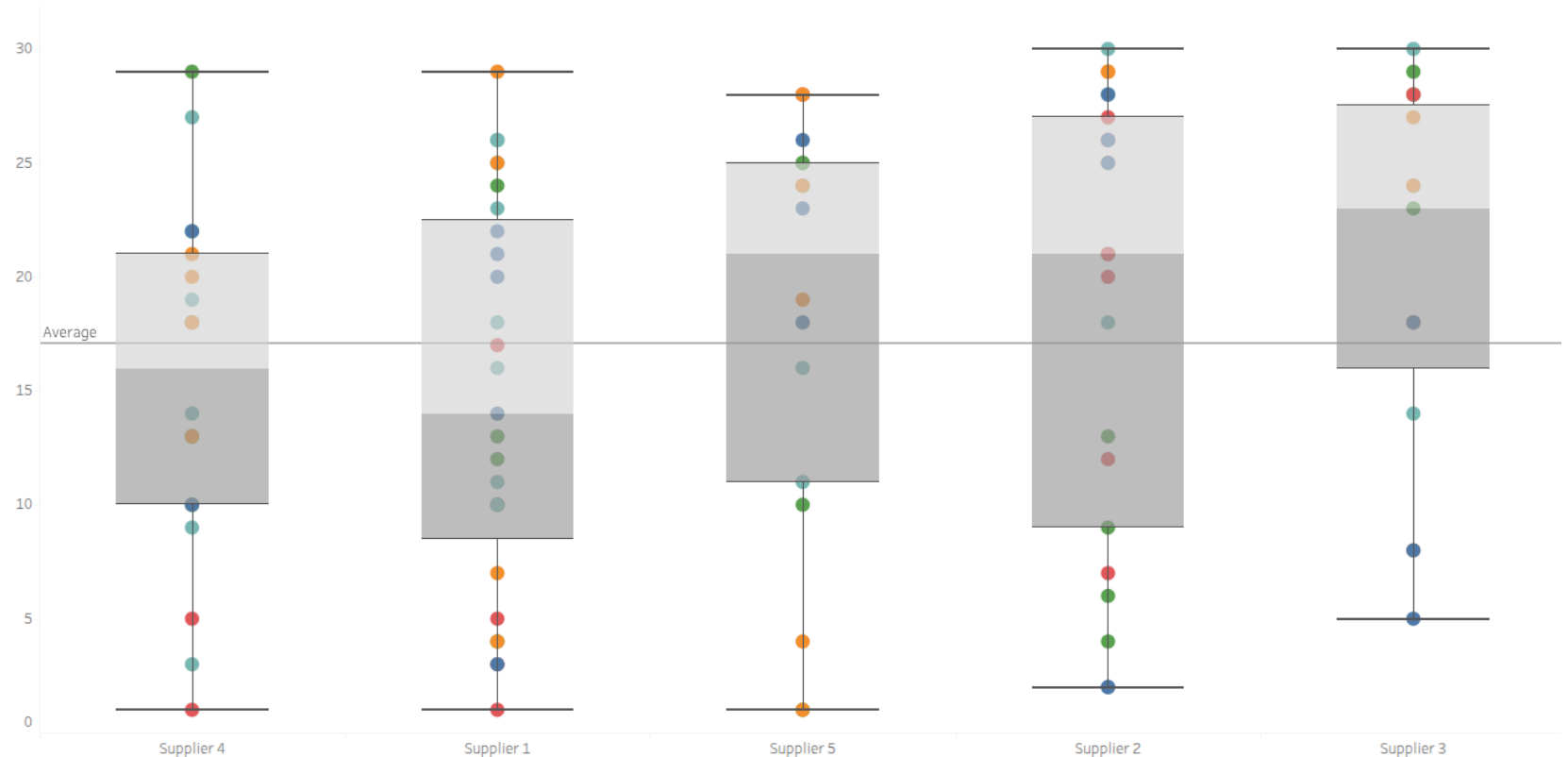


The Results

Supplier 3 presents the greatest fulfillment bottleneck

Insight:

When comparing lead times and defect rates across all suppliers, analysis reveals that Supplier 3 has an average delivery lead time of 20.13 days with a 2.47% defect rate, whereas Supplier 1 delivers faster with an average delivery lead time of 14.78 days and a lower defect rate of 1.80%.



The Impact

Quality improvements in Haircare and Skincare lines will yield large gains

Insight:

Because nearly \$4 out of every \$5 spent on manufacturing go toward Haircare and Skincare, even minor quality improvements in these two product lines will yield large operational gains.

Bringing the average defect rate of Haircare (2.48%) and Skincare (2.33%) down below the AQL threshold of 2.0% will:

- Eliminate approximately 214 defective units across production runs
- Yield approximately \$10,470 in direct manufacturing cost savings per cycle by preventing scrapped raw materials and rework labor
- Restore sellable inventory volume without increasing raw material inputs or plant operating hours
- Speed up overall batch approval turnaround by reducing the volume of inventory stuck in quality control holds